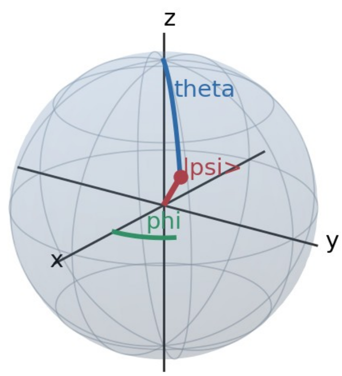
THE BLOCH SPHERE

A Geometric Language for Single-Qubit States

From normalized state vectors and global phase to rotations, measurement, tomography, noise, QKD, and density matrices

Bloch-sphere coordinates: polar angle theta and azimuth phi



Quantum Mathematics Roadmap - Topic 13

Prepared as a standalone reference volume

Contents

How to use this volume: Sections 1-7 establish the representation. Sections 8-12 turn algebraic gate and measurement rules into geometry. Sections 13-16 connect the sphere to experimental practice, noise, QKD, and multi-qubit limitations. The worked examples and exercises are designed for active revision.

Learning objectives

- Derive the Bloch-sphere angles from a normalized one-qubit state.
- Convert between amplitudes, density matrices, Pauli expectations, and Bloch vectors.
- Predict the geometric action of common single-qubit gates.
- Compute arbitrary-axis measurement probabilities using dot products.
- Interpret tomography and common noise channels geometrically.
- Use the sphere correctly in BB84 and recognize where it fails for multi-qubit systems.

1. Why a sphere can represent a qubit

A pure qubit is written as |psi> = alpha|0> + beta|1>, where alpha and beta are complex amplitudes. At first sight this appears to require four real numbers: the real and imaginary parts of alpha and beta. Physical constraints remove two of those real degrees of freedom. Normalization removes one, and global phase removes another. Exactly two real parameters remain, so a two-dimensional surface is sufficient.

$$|\alpha|^2 + |\beta|^2 = 1$$

The Bloch sphere is that two-parameter surface. Every pure one-qubit state corresponds to one point on the unit sphere. It does not depict a tiny physical ball inside the qubit. It is a coordinate model for the state space after physically redundant information has been removed.

Central idea: The sphere is not an extra assumption. It is the geometry already hidden inside the normalized two-component complex state vector.

1.1 Degrees of freedom count

Starting description	Constraint or equivalence	Real parameters remaining
alpha, beta in C	none	4
normalized state	|alpha|^2+|beta|^2=1	3
physical pure state	global phases e^(i gamma) identified	2

1.2 Why this geometry matters

- Basis states become directions rather than abstract column vectors.
- Unitary gates become rotations, making composition and control intuitive.
- Measurement probabilities become projections onto axes.
- Coherence and mixedness become distance from the center of the Bloch ball.
- Noise channels become contractions, translations, or distortions of vectors.

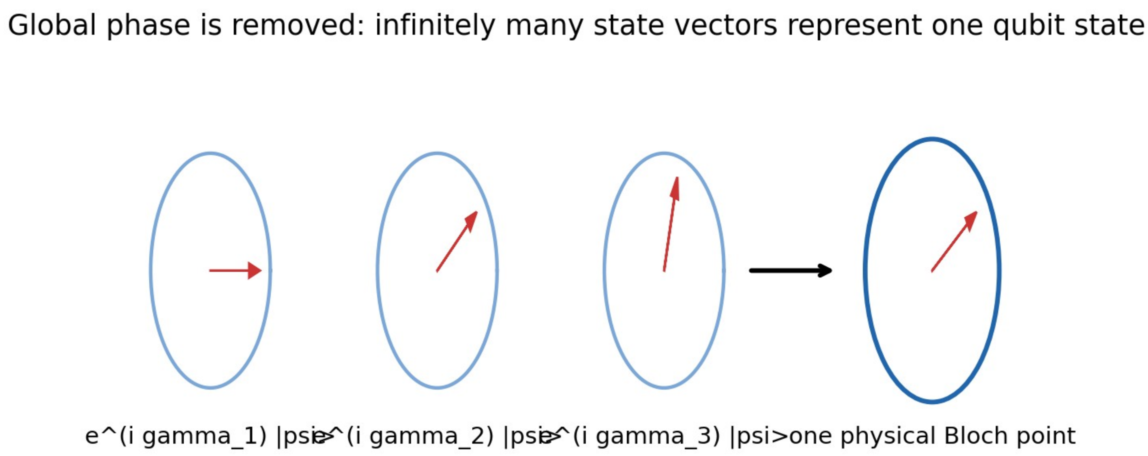


Figure 1. Global phases are physically equivalent and collapse to the same Bloch-sphere point.

2. Deriving the Bloch-sphere parameterization

2.1 Polar form of the amplitudes

Write each amplitude in magnitude-phase form: alpha = r0 e^(i a) and beta = r1 e^(i b), with r0,r1 >= 0. Normalization gives r0^2+r1^2=1. Therefore there is an angle theta in [0,pi] such that r0=cos(theta/2) and r1=sin(theta/2).

$$|\psi\rangle = e^{i a} \cos(\theta/2)|0\rangle + e^{i b} \sin(\theta/2)|1\rangle$$

Factor out e^(i a). This common factor is a global phase, so it has no effect on any isolated-state measurement probability. Let phi=b-a be the relative phase.

$$|\psi\rangle \sim \cos(\theta/2)|0\rangle + e^{i \phi} \sin(\theta/2)|1\rangle$$

The symbol ~ means physically equivalent up to global phase. The conventional Bloch-sphere representation is therefore

$$|\psi(\theta,\phi)\rangle = \cos(\theta/2)|0\rangle + e^{i \phi} \sin(\theta/2)|1\rangle$$

2.2 Why the half-angle appears

Measurement probabilities are squared magnitudes. If theta itself appeared directly in the amplitudes, the north-to-south sweep would not align with the standard polar coordinate. The half-angle ensures that theta=0 gives |0>, theta=pi gives |1>, and theta=pi/2 gives equal computational-basis probabilities.

theta	State form	Location
0	|0>	north pole
pi/2	(|0>+e^(i phi)|1>)/sqrt(2)	equator

theta	State form	Location
pi	e^(i phi)|1> ~ |1>	south pole

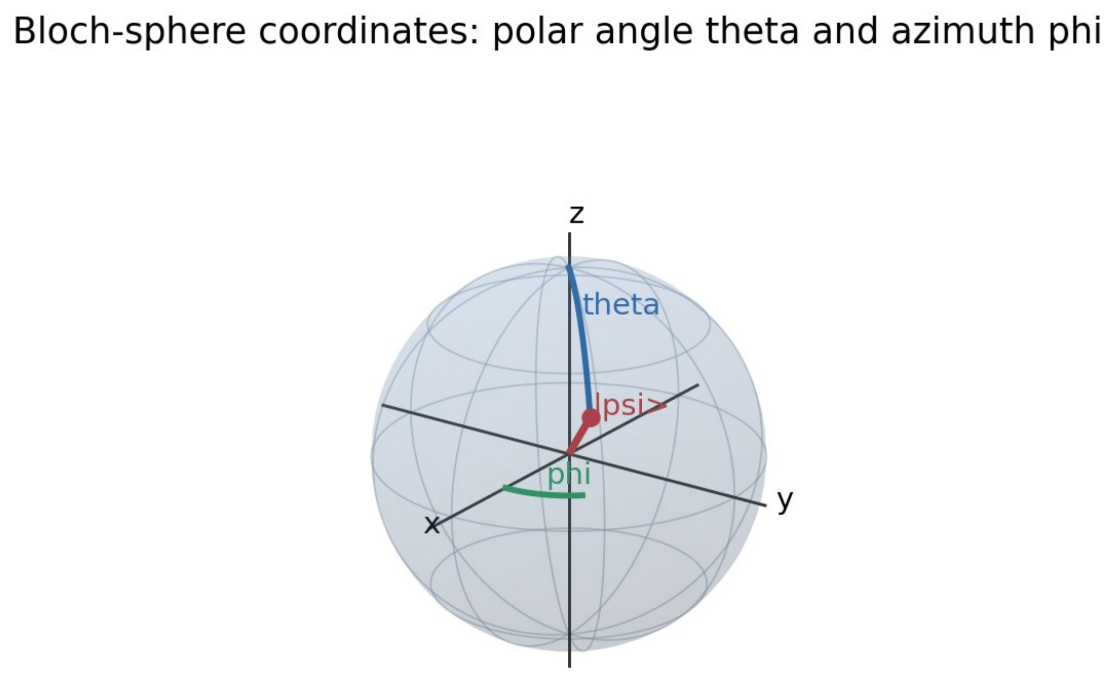


Figure 2. The point is specified by polar angle theta and azimuth phi.

3. Cartesian Bloch coordinates

The point associated with |psi(theta,phi)> has Cartesian coordinates

$$x = \sin(\theta)\cos(\phi), \quad y = \sin(\theta)\sin(\phi), \quad z = \cos(\theta)$$

These satisfy x^2+y^2+z^2=1 for every pure state. The vector r=(x,y,z) is called the Bloch vector. The reverse conversion is theta=arccos(z) and phi=atan2(y,x), with phi irrelevant at either pole.

3.1 Directly from amplitudes

For a normalized state alpha|0>+beta|1>, the Bloch coordinates can be extracted without first finding theta and phi:

$$x = 2\,\mathrm{Re}(\alpha^* \beta), \quad y = 2\,\mathrm{Im}(\alpha^* \beta), \quad z = |\alpha|^2 - |\beta|^2$$

Sign convention: Here alpha* beta means the complex conjugate of alpha multiplied by beta. With the standard Pauli Y matrix, y=2 Im(alpha* beta). Some density-matrix formulas display the equivalent relation y=-2 Im(rho01) because rho01=alpha beta*.

3.2 From coordinates back to amplitudes

Choose theta=arccos(z) and phi=atan2(y,x). Then a canonical representative is cos(theta/2)|0>+e^(i phi)sin(theta/2)|1>. Any common phase may be multiplied onto this vector without changing the Bloch point.

3.3 Example

Let |psi>=(sqrt(3)/2)|0>+(i/2)|1>. Then alpha=sqrt(3)/2 and beta=i/2. The coordinates are x=0, y=sqrt(3)/2, z=1/2. Thus theta=pi/3 and phi=pi/2.

$$r = (0, \mathrm{sqrt}(3)/2, 1/2)$$

Geometric check: The vector has unit length: 0^2 + 3/4 + 1/4 = 1.

4. Landmark states and mutually unbiased bases

The eigenstates of the three Pauli operators define six special directions. Opposite points correspond to orthogonal states. The x, y, and z axes form three mutually unbiased measurement bases: knowing a state exactly in one axis gives uniformly random outcomes when measured along either perpendicular axis.

Bloch direction	Ket	Pauli eigenvalue
+z	|0>	Z: +1
-z	|1>	Z: -1
+x	|+>=(|0>+|1>)/sqrt(2)	X: +1
-x	|->=(|0>-|1>)/sqrt(2)	X: -1
+y	|+i>=(|0>+i|1>)/sqrt(2)	Y: +1
-y	|-i>=(|0>-i|1>)/sqrt(2)	Y: -1

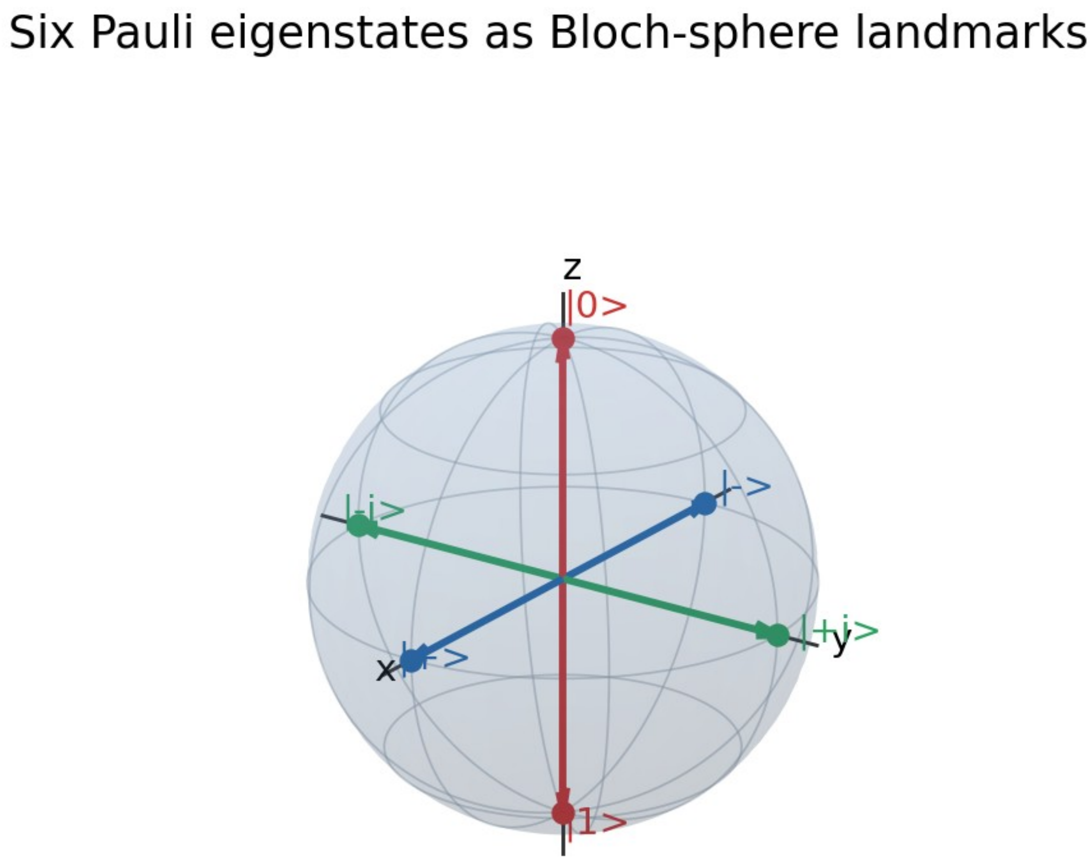


Figure 3. The six eigenstates of X, Y, and Z occupy the axis endpoints.

4.1 Orthogonality as antipodality

For pure qubits, two states are orthogonal exactly when their Bloch vectors point in opposite directions. For example, |0> and |1> are antipodal, as are |+> and |->. This statement is special to two-dimensional pure-state space.

$$|\langle psi|phi\rangle|^2 = (1 + r \text{ dot } s)/2$$

If s=-r, then the overlap is zero. If s=r, the overlap is one. If r dot s=0, the states have squared overlap 1/2 and are mutually unbiased.

5. Global phase and relative phase

5.1 Global phase

Multiplying the entire state by e^(i gamma) changes neither normalization nor any probability. The rays |psi> and e^(i gamma)|psi> therefore represent the same physical pure state. This is why the state space is projective rather than an ordinary unit sphere in C^2.

$$|\langle m|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}\langle m|\psi\rangle|^2 = |\langle m|\psi\rangle|^2$$

5.2 Relative phase

The phase between the |0> and |1> amplitudes changes the Bloch azimuth phi and can alter interference. Equal computational-basis probabilities do not imply the same state: every point on the equator has P(0)=P(1)=1/2, yet different equatorial states respond differently to X- and Y-basis measurements.

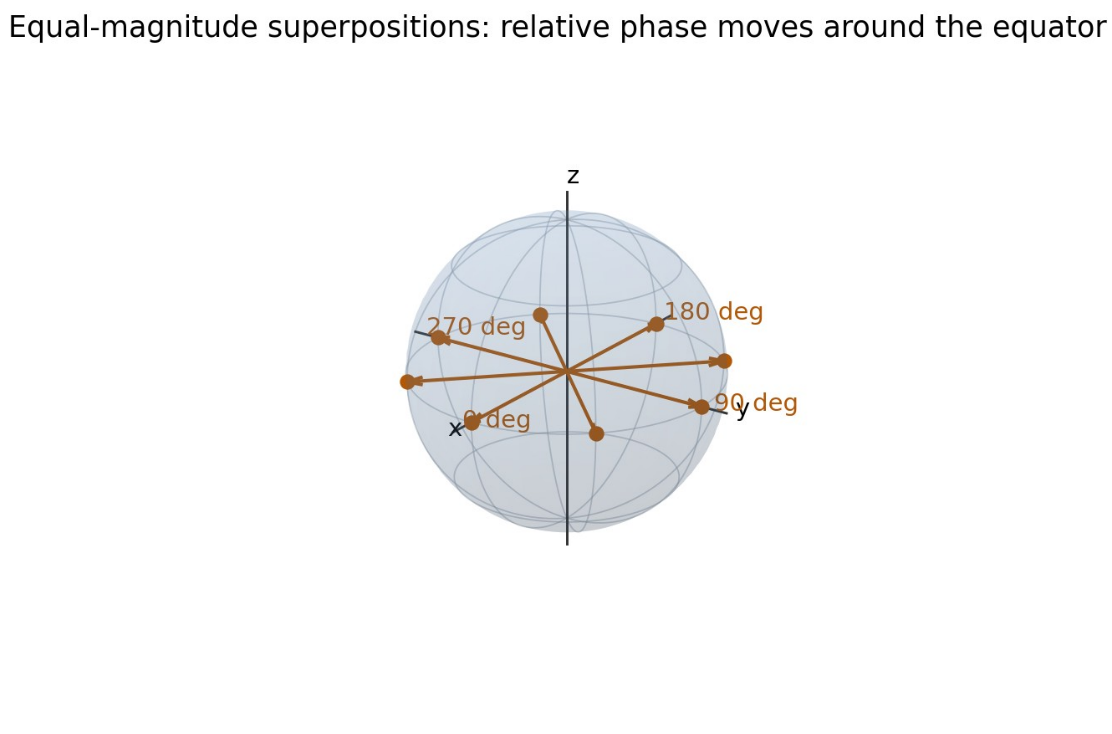


Figure 4. Relative phase moves an equal-magnitude superposition around the equator.

5.3 Making phase observable

A computational-basis measurement reads only z. To reveal azimuthal phase, rotate the desired transverse axis onto z before measurement. For example, applying H before a Z measurement performs an X-basis measurement. Applying S^dagger followed by H performs a Y-basis measurement.

Interference principle: Relative phase becomes measurable only when alternatives are recombined so that their amplitudes add or cancel.

6. Probabilities, expectation values, and observables

6.1 Computational-basis probabilities

For |psi(theta,phi)>, the Z-basis probabilities are

$$P(0)=\cos^2(\theta/2)=(1+z)/2, \quad P(1)=\sin^2(\theta/2)=(1-z)/2$$

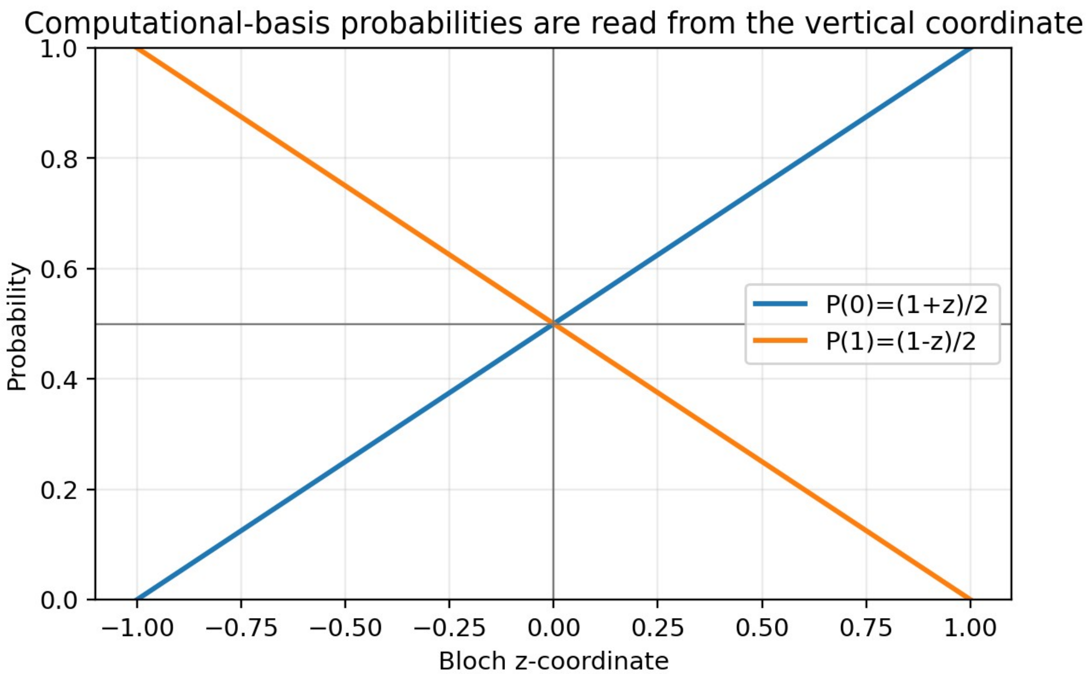


Figure 5. The vertical coordinate determines computational-basis probabilities.

6.2 Pauli expectation values

The Bloch coordinates are exactly the expectation values of the Pauli operators:

$$x=\langle X\rangle, \quad y=\langle Y\rangle, \quad z=\langle Z\rangle$$

This gives the operational meaning of r. It is not merely a drawing coordinate; its components are experimentally estimable averages of +/-1 outcomes.

6.3 General Hermitian observable

Every one-qubit Hermitian matrix can be written as A=a0 I + a dot sigma, where sigma=(X,Y,Z) and a is a real three-vector. Its expectation value in state r is

$$\langle A\rangle = a_0 + a \cdot r$$

Thus single-qubit expectation values reduce to Euclidean dot products. The observable eigenvalues are a0 +/- |a|, and its eigenstates lie along +/-a.

6.4 Variance of a Pauli-direction measurement

$$\mathrm{Var}(n \text{ dot sigma}) = 1 - (n \text{ dot } r)^2$$

A state aligned with n has zero variance; a state perpendicular to n produces equally likely +/- outcomes and maximal variance one.

7. Density matrices and the Bloch ball

7.1 Bloch decomposition

A general one-qubit density matrix has the unique form

$$\text{rho} = (I + r \text{ dot sigma})/2 = (I + xX + yY + zZ)/2$$

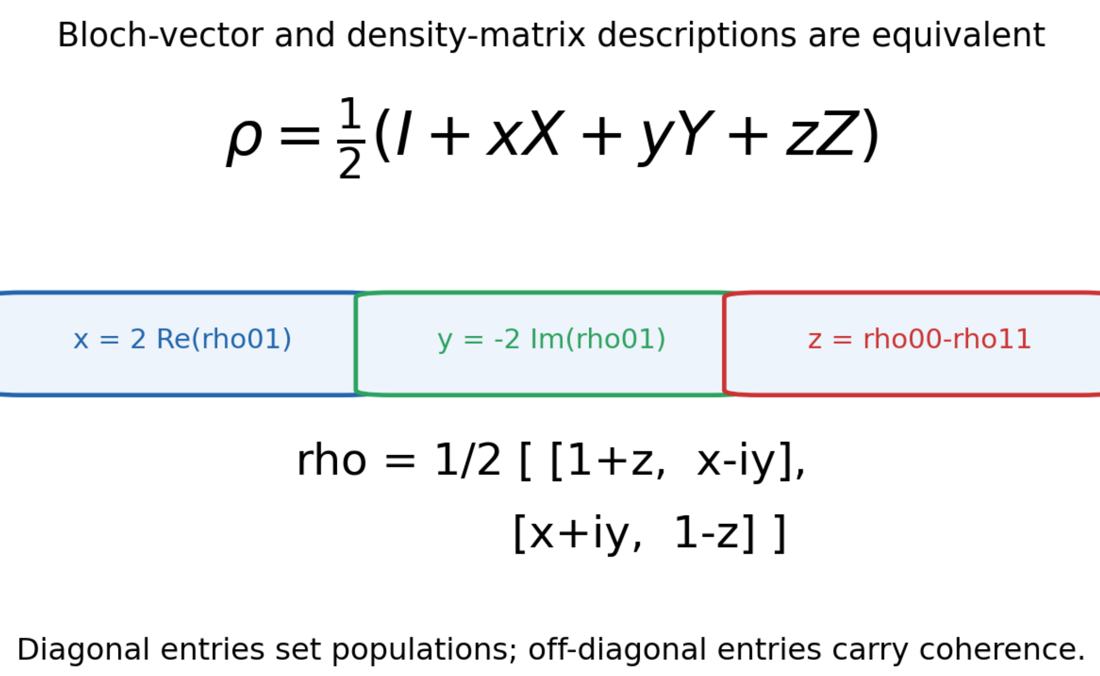


Figure 6. Density-matrix entries and Bloch coordinates encode the same information.

Written explicitly,

$$\text{rho} = [[(1+z)/2, (x-iy)/2], [(x+iy)/2, (1-z)/2]]$$

7.2 Positivity and the unit ball

The eigenvalues of rho are $(1+|r|)/2$ and $(1-|r|)/2$. Positivity therefore requires $|r|<=1$. Pure states have $|r|=1$ and lie on the surface. Mixed states have $|r|<1$ and occupy the interior. The maximally mixed state I/2 lies at the center.

$$\mathrm{Tr}(\text{rho}^2) = (1 + |r|^2)/2$$

Purity ranges from 1/2 at the center to 1 on the surface. The radial length is therefore a direct measure of single-qubit purity.

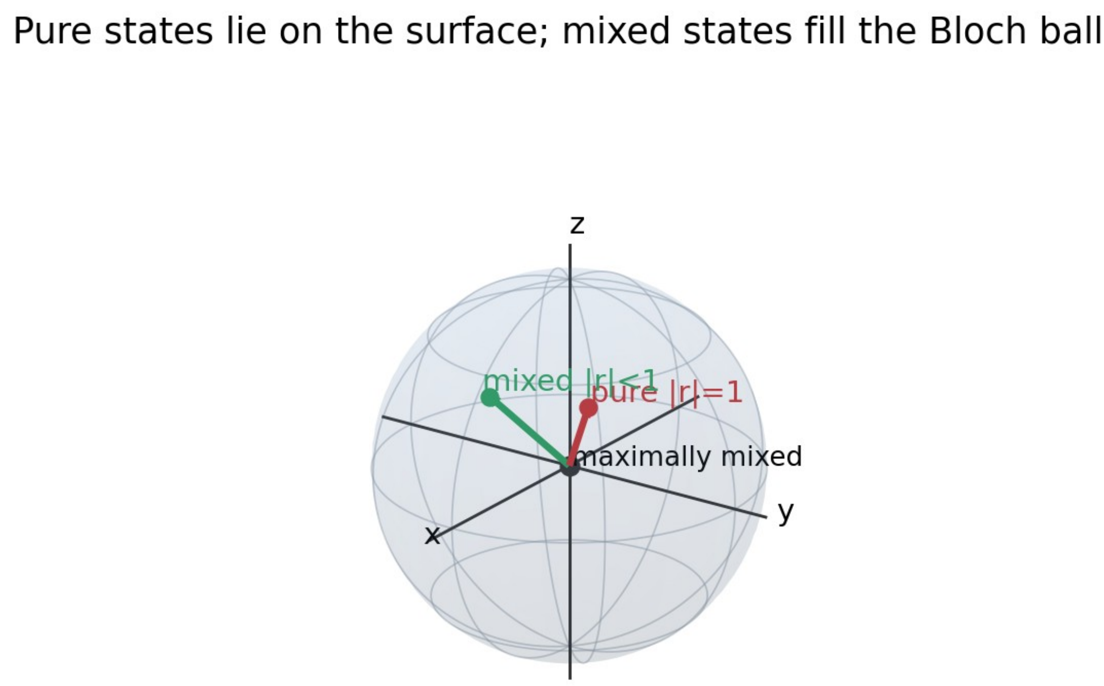


Figure 7. Pure states form the surface; mixed states fill the interior.

7.3 Populations and coherence

The diagonal entries rho00 and rho11 are computational-basis populations. The off-diagonal entries are coherences. In Bloch coordinates, z records the population imbalance, while x and y encode the real and imaginary parts of coherence.

Quantity	**Density-matrix expression**	**Bloch meaning**
Population of |0>	rho00=(1+z)/2	northward bias
Population of |1>	rho11=(1-z)/2	southward bias
Real coherence	Re(rho01)=x/2	x component
Imaginary coherence	Im(rho01)=-y/2	y component
Purity	Tr(rho^2)	radial length

7.4 Classical mixture versus coherent superposition

The pure state |+> and the equal classical mixture of |0> and |1> both produce 50-50 outcomes in the Z basis. They are nevertheless different. |+> has Bloch vector (1,0,0), whereas the mixture has r=(0,0,0). X-basis measurement distinguishes them immediately.

$$|+\rangle\langle+| = (I+X)/2, \quad 0.5|0\rangle\langle 0|+0.5|1\rangle\langle 1| = I/2$$

Key warning: A probability distribution in one basis does not completely specify a quantum state. Coherence requires measurements in complementary bases.

7.5 Convex mixtures

If rho=sum_k p_k rho_k, then its Bloch vector is the weighted average r=sum_k p_k r_k. Convex mixing therefore produces points inside the ball by averaging surface points. Different ensembles can yield the same interior density matrix, so the decomposition into pure states is generally not unique.

8. Quantum gates as spatial rotations

A one-qubit unitary transforms rho as rho -> U rho U^dagger. In Bloch form this becomes r -> R r, where R is a real three-dimensional rotation matrix with determinant +1. Global phase in U disappears, so physically distinct single-qubit unitaries correspond to rotations of the Bloch sphere.

U (r dot sigma) U^dagger = (Rr) dot sigma

8.1 Axis-angle form

A rotation through Bloch angle theta about unit axis n is generated by

R_n(theta) gate = exp[-i theta (n dot sigma)/2]

The factor 1/2 is essential: the SU(2) unitary parameter uses half the physical Bloch rotation angle. This is the same half-angle structure seen in the state parameterization.

8.2 Rodrigues formula for the Bloch vector

r' = r cos(theta) + (n x r) sin(theta) + n(n dot r)(1-cos(theta))

This is the ordinary three-dimensional rotation formula. It provides a fast way to predict gate action without multiplying state vectors.

8.3 Rotations preserve purity and angles

Because R is orthogonal, |Rr|=|r| and (Rr) dot (Rs)=r dot s. Unitary gates preserve purity, state overlaps, and distinguishability. Noise channels, in contrast, may shrink or translate the ball.

Global phase of a gate: Multiplying U by e^(i gamma) leaves the associated Bloch rotation unchanged. U(2) therefore reduces to SO(3) after global phase is ignored.

9. Pauli gates and elementary rotations

The Pauli gates are pi rotations about their corresponding axes, up to a global phase. Their Bloch actions are especially simple.

Gate	Bloch rotation	Coordinate action
X	pi about x	(x,y,z)->(x,-y,-z)
Y	pi about y	(x,y,z)->(-x,y,-z)
Z	pi about z	(x,y,z)->(-x,-y,z)

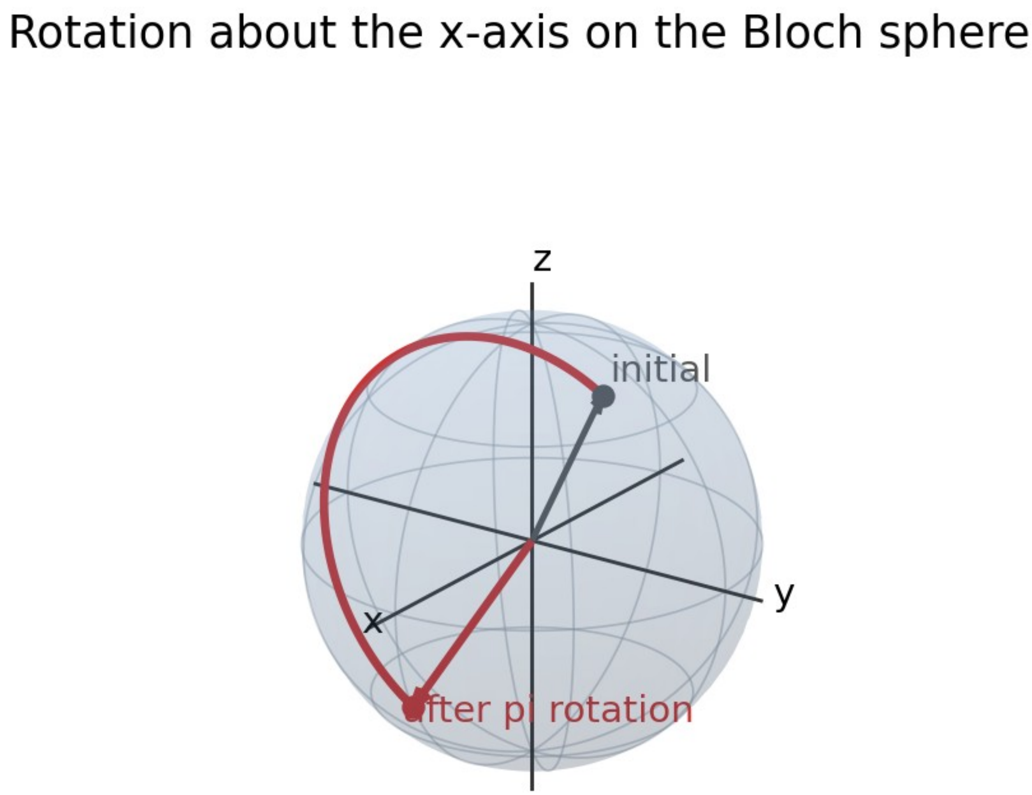


Figure 8. X rotates the Bloch vector by pi about the x-axis.

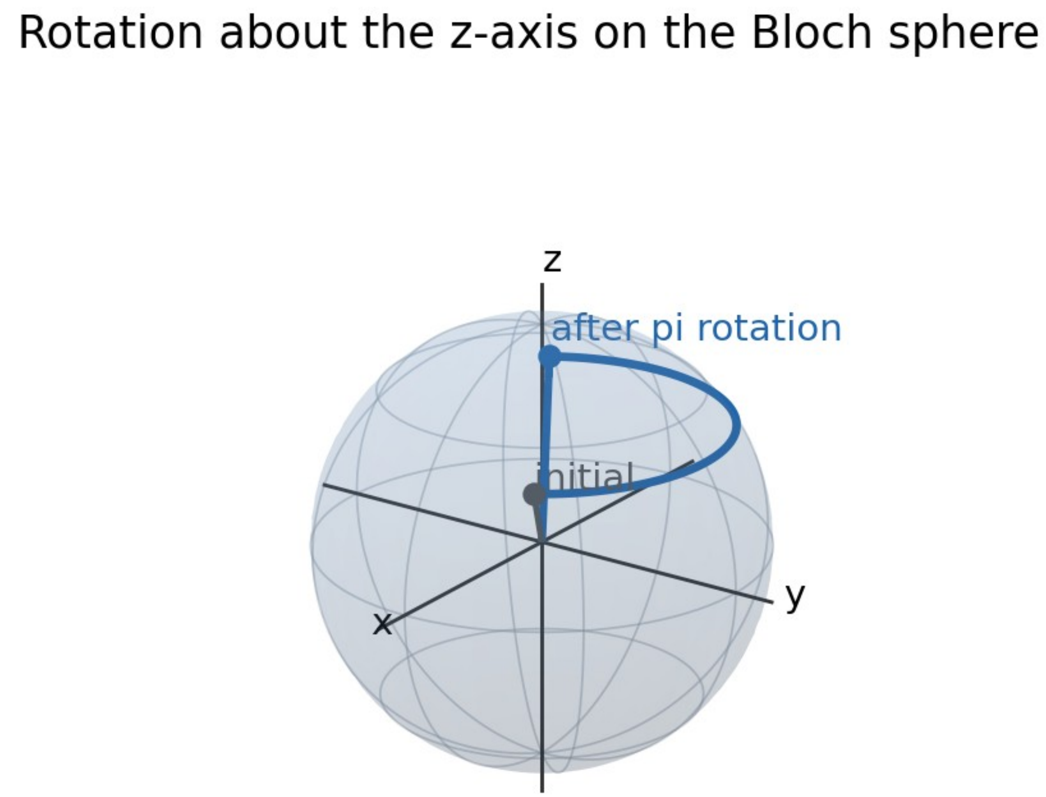


Figure 9. Z rotates the Bloch vector by pi about the z-axis.

9.1 Basis-state action

- X swaps |0> and |1>, so it exchanges north and south poles.
- Z leaves |0> and |1> at their poles but swaps |+> and |->.
- Y also swaps the poles, with relative phases that are invisible as global phases for basis states but important in superpositions.

10. Hadamard, phase, and rotation gates

10.1 Hadamard

H is Hermitian and satisfies H^2=I. Its Bloch action exchanges x and z while reversing y:

$$(x,y,z) \to (z,-y,x)$$

Geometrically this is a pi rotation about the axis (x+z)/sqrt(2), not a 90-degree rotation. The familiar mapping |0>->|+> follows because +z is rotated to +x.

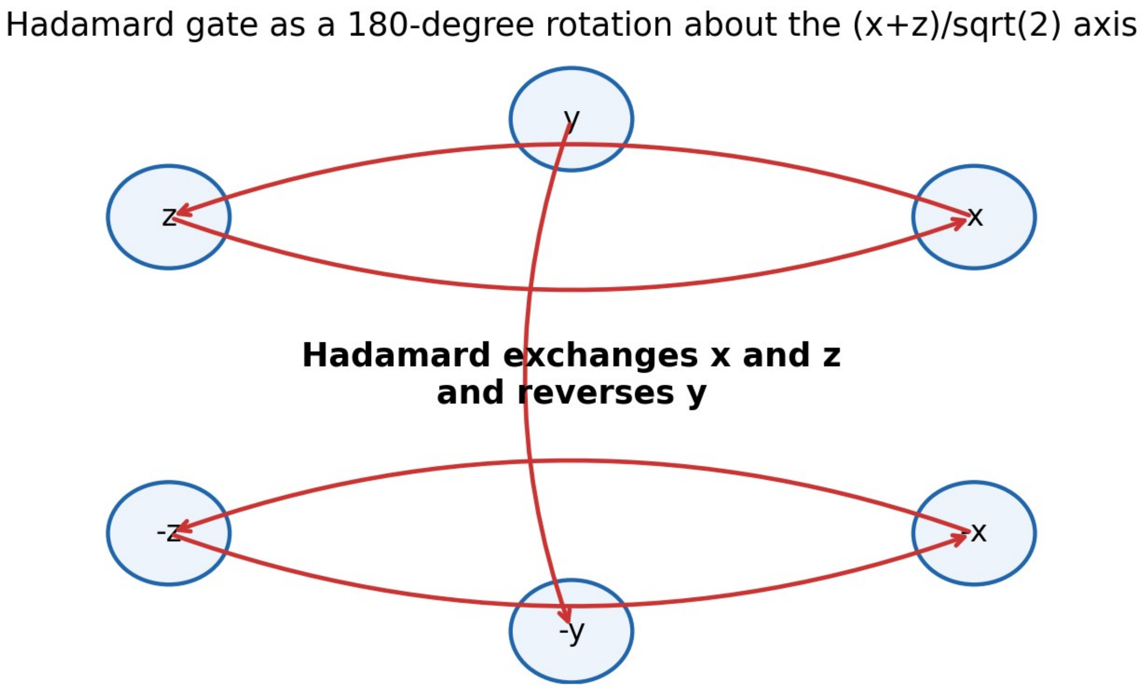


Figure 10. Hadamard exchanges the computational and X bases.

10.2 Phase gates

The phase gate P(lambda)=diag(1,e^(i lambda)) is equivalent up to global phase to Rz(lambda). It rotates the equator by lambda while leaving z unchanged. S=P(pi/2) and T=P(pi/4).

$$P(\lambda) \sim Rz(\lambda) = \exp(-i\, \lambda\, Z/2)$$

Gate	Azimuth change	Example
S	phi -> phi+pi/2	|+> -> |+i>
T	phi -> phi+pi/4	moves +x halfway toward +y
Z	phi -> phi+pi	|+> -> |->

10.3 Rx, Ry, Rz

$$Rx(\theta)=\cos(\theta/2)I-i\,\sin(\theta/2)X$$

$$Ry(\theta)=\cos(\theta/2)I-i\,\sin(\theta/2)Y$$

$$Rz(\theta)=\cos(\theta/2)I-i\,\sin(\theta/2)Z$$

These gates are the basic continuous rotations used to prepare arbitrary qubit states and compile general one-qubit unitaries.

11. Arbitrary-axis rotations and the SU(2) double cover

11.1 Arbitrary axis

For a unit vector n=(nx,ny,nz), define n dot sigma=nxX+nyY+nzZ. Since (n dot sigma)^2=I, the exponential simplifies exactly:

$$U_n(\theta)=\cos(\theta/2)I - i \sin(\theta/2)(n \cdot \sigma)$$

The eigenstates along +/-n acquire opposite phases under this unitary. Every special unitary one-qubit gate can be written in this axis-angle form.

11.2 Why a 2pi rotation changes the state vector sign

$$U_n(2\pi)= -I, \quad U_n(4\pi)=I$$

A 2pi Bloch rotation returns every point to itself but multiplies each state vector by -1. For an isolated qubit that sign is global and unobservable. In a controlled or interferometric setting, however, the relative sign can become observable. This two-to-one relationship is called the double cover SU(2)->SO(3).

Do not confuse: The Bloch sphere returns to the same point after 2pi, while the spinor representative returns exactly only after 4pi. Both statements are correct because the sphere already identifies global phase.

11.3 Euler decomposition

Any one-qubit unitary, up to global phase, can be decomposed as three axis rotations, for example

$$U \sim R_z(\alpha) R_y(\beta) R_z(\gamma)$$

On the sphere this is simply a sequence of ordinary rotations. The noncommutativity of gates is the noncommutativity of three-dimensional rotations about different axes.

12. Measurement geometry

12.1 Measurement along an arbitrary axis

The observable n dot sigma has eigenvalues +1 and -1, with projectors

$$P_+ = (I+n \cdot \sigma)/2, \quad P_- = (I-n \cdot \sigma)/2$$

For state rho=(I+r dot sigma)/2, the Born rule becomes

$$p(+)=(1+n \cdot r)/2, \quad p(-)=(1-n \cdot r)/2$$

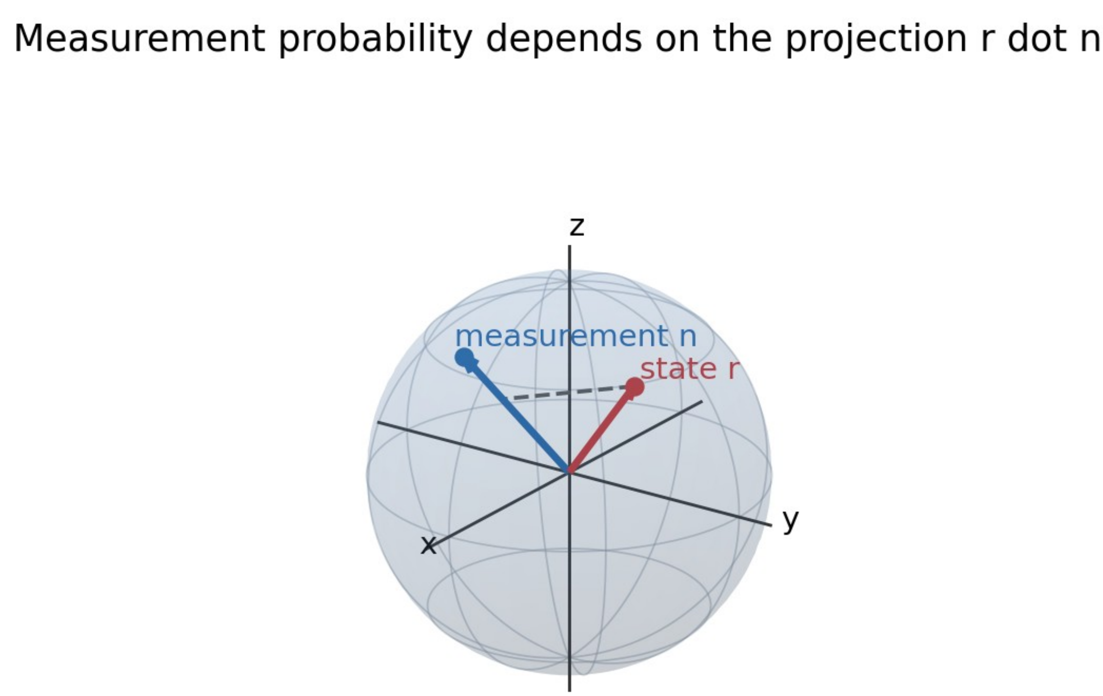


Figure 11. The dot product with the measurement axis determines the outcome bias.

12.2 Geometric interpretation

- If r=n, the + outcome is certain.
- If r=-n, the - outcome is certain.
- If r is perpendicular to n, both outcomes occur with probability 1/2.
- For mixed states, reduced length means reduced bias for every axis.

12.3 State update

An ideal rank-one projective measurement sends the post-measurement Bloch vector to $+n$ or $-n$ according to the observed result. The original vector is not merely revealed; it is generally replaced by an eigenstate of the measured observable.

Measurement disturbance: A measurement changes the state unless the pre-measurement vector is already aligned or anti-aligned with the measurement axis.

13. Single-qubit state tomography

Tomography estimates an unknown density matrix from repeated measurements on identically prepared systems. Because $\rho=(I+xX+yY+zZ)/2$, estimating $\langle X\rangle$, $\langle Y\rangle$, and $\langle Z\rangle$ is sufficient.

$$\hat{r} = (\widehat{\langle X\rangle}, \widehat{\langle Y\rangle}, \widehat{\langle Z\rangle})$$

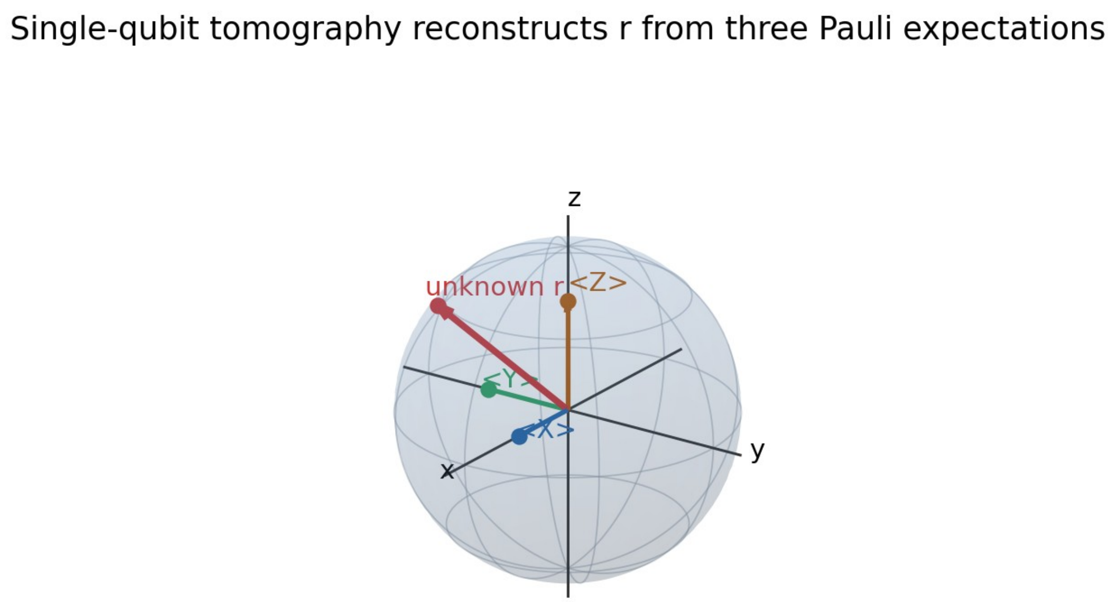


Figure 12. Measuring the three Pauli axes reconstructs the Bloch vector.

13.1 Experimental estimator

Suppose N_{x+} and N_{x-} are the counts for +/- outcomes in an X-basis experiment. Then

$$\widehat{\langle X\rangle} = (N_{x+} - N_{x-}) / (N_{x+} + N_{x-})$$

Analogous formulas estimate y and z. Finite-shot fluctuations can produce an unphysical estimate with $|\hat{r}|>1$. Practical reconstruction methods may project the estimate back into the Bloch ball or use maximum-likelihood estimation to enforce positivity.

13.2 Measurement circuits

Desired expectation	**Pre-rotation before Z measurement**	**Reason**
$\langle Z\rangle$	none	already computational basis
$\langle X\rangle$	H	$H X H = Z$
$\langle Y\rangle$	$S^\dagger$ then H	maps Y eigenstates to Z basis

13.3 Sample complexity

Each estimated component is an average of bounded ± 1 variables. Its standard error scales as $1/\sqrt{N}$, where N is the number of shots for that basis. Doubling precision therefore requires roughly four times as many shots.

Tomography is destructive: The same physical qubit cannot be measured along all three axes. Tomography requires many independently prepared copies of the same state.

14. Distance, fidelity, and distinguishability

14.1 Pure-state overlap

For pure qubits with unit Bloch vectors r and s,

$$F = |\langle\psi|\phi\rangle|^2 = (1+r \cdot s)/2 = \cos^2(\Delta/2)$$

Here Delta is the angle between r and s. Nearby points have high fidelity; antipodal points have zero fidelity.

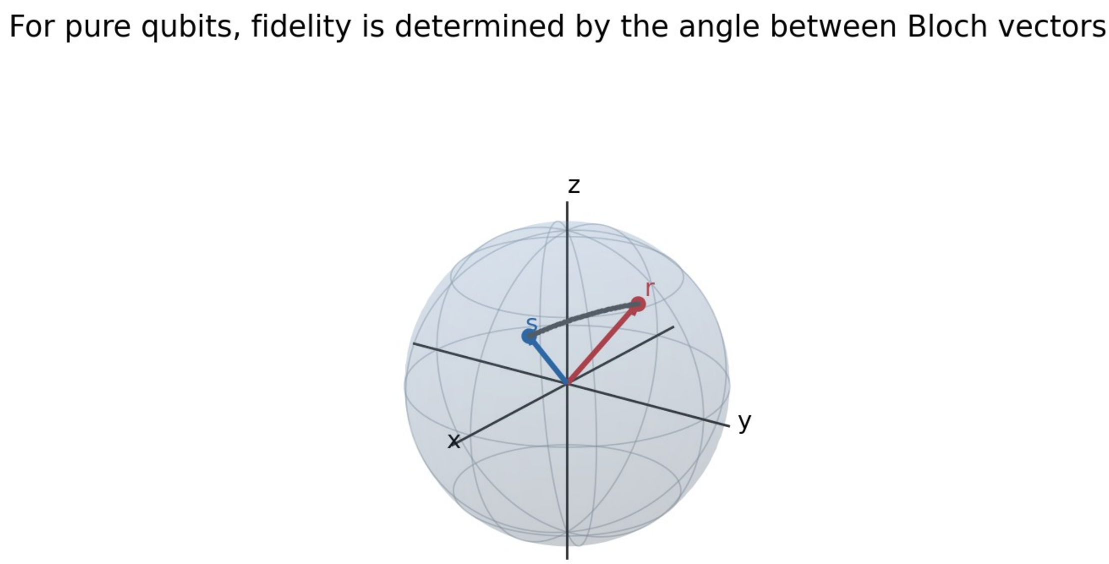


Figure 13. Pure-state fidelity is controlled by angular separation.

14.2 Trace distance for qubits

For two qubit density matrices rho and sigma with Bloch vectors r and s, the trace distance is

$$D(rho,sigma) = (1/2)|r-s|$$

This has a direct operational meaning: with equal prior probabilities, the best single-shot discrimination success probability is (1+D)/2.

14.3 Mixed-state fidelity

Using the squared-fidelity convention, one-qubit states satisfy

$$F(rho,sigma)= [1+r \text{ dot } s + sqrt((1-|r|^2)(1-|s|^2))]/2$$

When both states are pure, the square-root term vanishes and the earlier angular formula is recovered.

14.4 Entropy from radial length

The eigenvalues of rho depend only on $|r|$, so the von Neumann entropy is a radial function. It is zero on the surface and one bit at the center.

$$S(rho)=h2((1+|r|)/2)$$

15. Noise channels on the Bloch ball

A general trace-preserving one-qubit channel acts affinely on Bloch vectors:

$$r \to A r + c$$

Unitary channels have c=0 and A a rotation. Noisy channels may shrink the ball, stretch it anisotropically, or translate its center. Complete positivity restricts which affine maps are physically allowed.

15.1 Dephasing

Phase damping suppresses x and y while leaving z unchanged. For a dephasing factor eta in [0,1],

$$(x,y,z) \to (\eta x, \eta y, z)$$

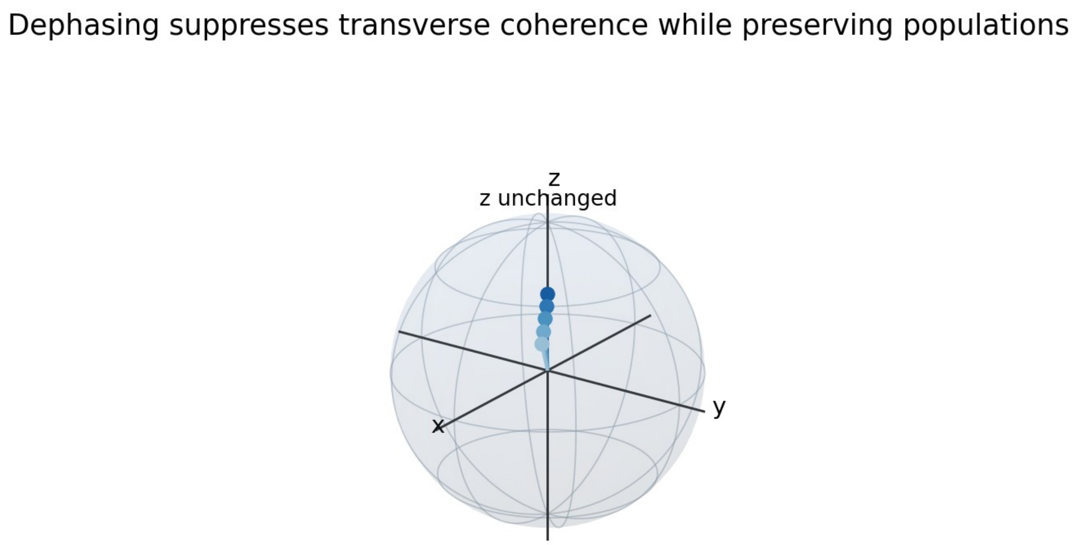


Figure 14. Dephasing erases transverse coherence but preserves Z-basis populations.

15.2 Depolarizing noise

A common convention maps rho -> (1-p)rho+pI/2, giving

$$r \to (1-p)r$$

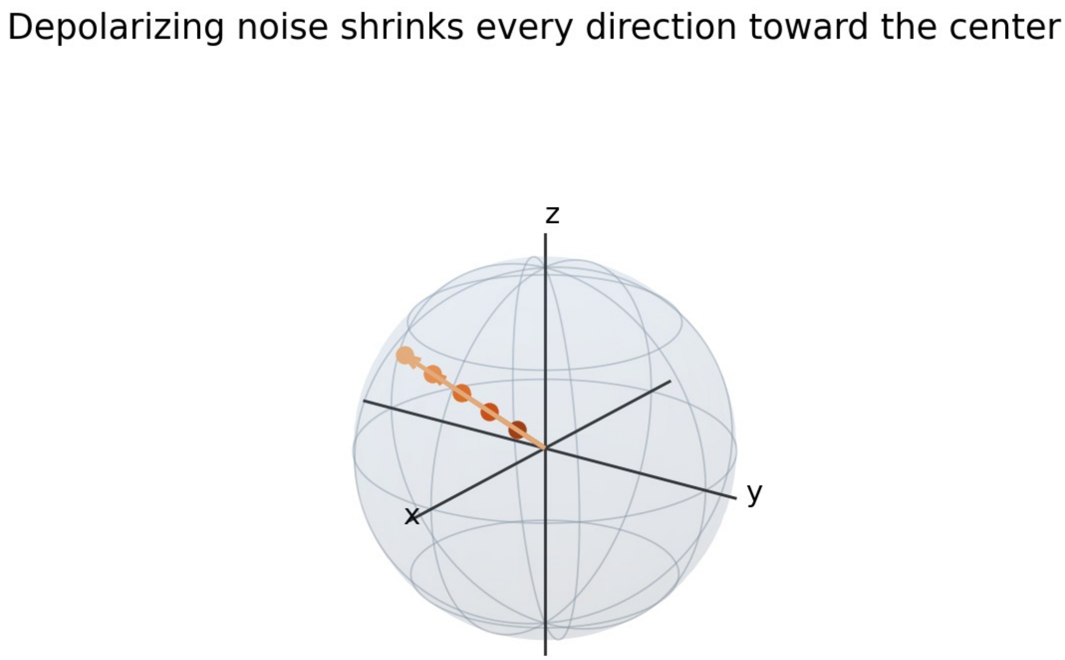


Figure 15. Depolarizing noise contracts all directions toward the maximally mixed state.

15.3 Amplitude damping

Amplitude damping models energy relaxation |1>->|0>. With damping probability gamma,

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(x,y,z) -> (sqrt(1-gamma)x, sqrt(1-gamma)y, (1-gamma)z+gamma)
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Unlike dephasing and depolarizing noise, this channel is non-unital: it moves the center of the Bloch ball toward +z. The ground state is the fixed point.

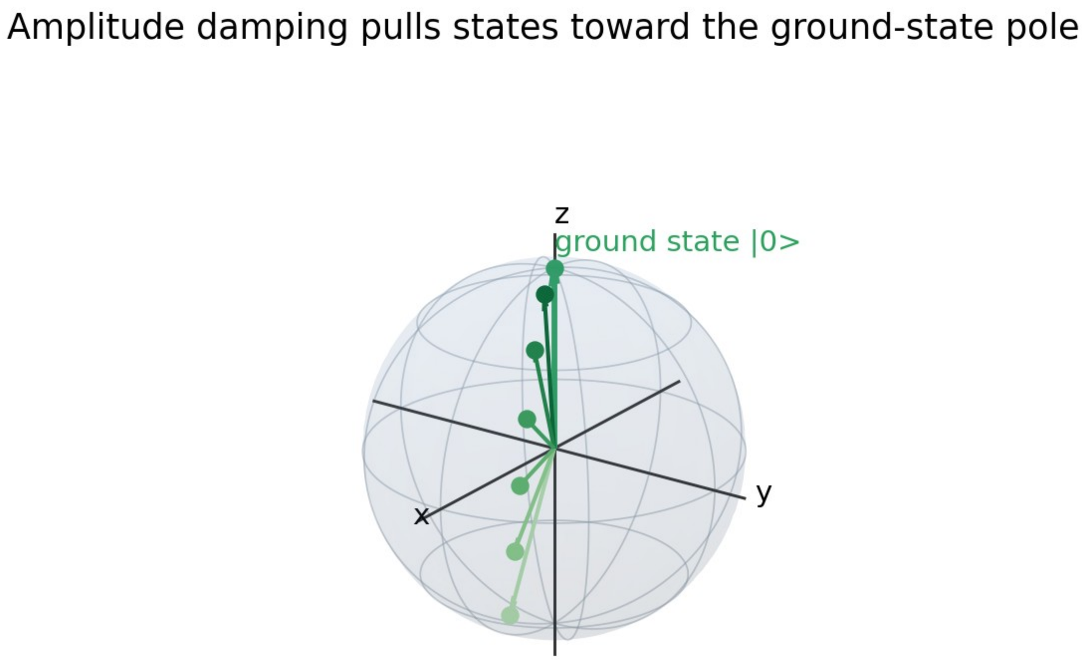

Figure 16. Relaxation contracts and translates vectors toward |0>.

15.4 Bit-flip and phase-flip channels

A probabilistic X error with probability p maps r to (x,(1-2p)y,(1-2p)z). A probabilistic Z error maps r to ((1-2p)x,(1-2p)y,z). Each preserves the component along its error axis while shrinking perpendicular components.

15.5 Coherence times

In many physical qubits, longitudinal relaxation is summarized by T1 and transverse coherence loss by T2. Idealized Markovian models give z(t) approaching its equilibrium value with e^(-t/T1), while x and y decay approximately as e^(-t/T2). Pure dephasing contributes an additional rate beyond relaxation.

$$1/T2 = 1/(2T1) + 1/T\_phi$$

Model limitation: The Bloch equations are powerful summaries, but real devices may show non-Markovian noise, leakage outside the qubit subspace, drift, and correlated errors that cannot be captured by a single stationary Bloch-ball map.

16. QKD states on the Bloch sphere

16.1 BB84

BB84 uses two mutually unbiased bases. The Z basis encodes bits in |0> and |1>; the X basis encodes bits in |+> and |->. On the sphere these are two perpendicular diameters.

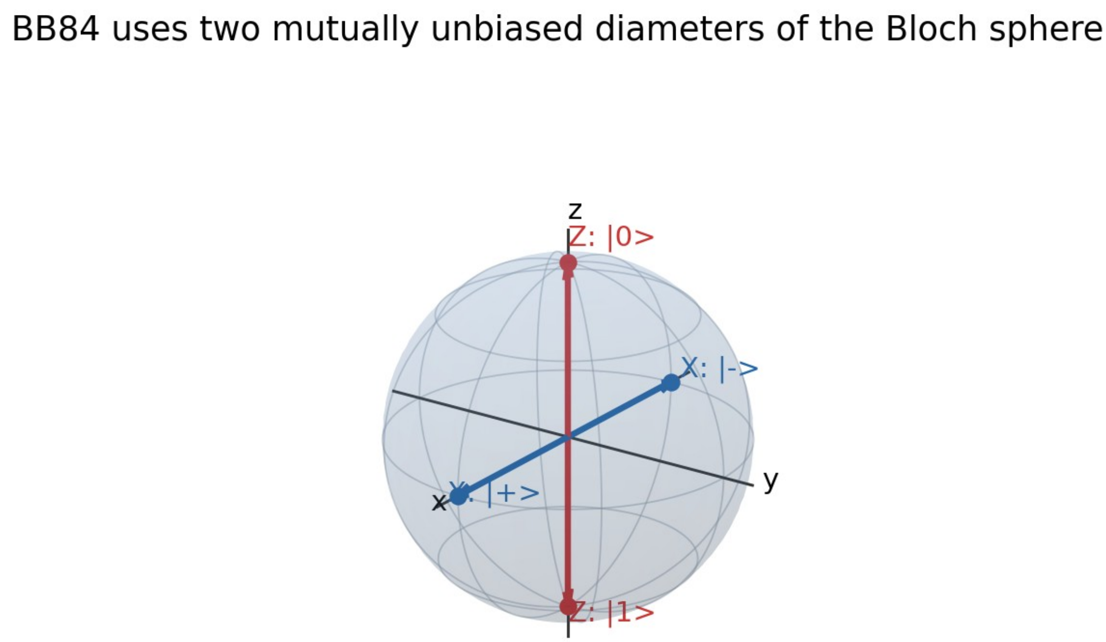

Figure 17. BB84 uses the z and x axes.

16.2 Why wrong-basis measurement is random

If a state lies along +/-z and is measured along x, the dot product is zero. Therefore p(+)=p(-)=1/2. The same holds when an X-basis state is measured in Z. This geometric orthogonality is the source of basis incompatibility used by BB84.

16.3 Intercept-resend attack

An eavesdropper who guesses the wrong basis projects the state onto the wrong diameter. When the legitimate receiver later uses the correct basis, the resent state yields a random result. In the ideal BB84 intercept-resend attack, this creates errors in a predictable fraction of the sifted key.

16.4 Six-state protocol

The six-state protocol uses all three Pauli bases, corresponding to the six axis endpoints. It probes noise more symmetrically and provides more information about the channel, at the cost of lower basis-matching efficiency.

16.5 Channel visualization

QKD channel noise can be interpreted as moving or shrinking the intended signal states. Dephasing mainly damages equatorial encodings, while amplitude damping introduces an asymmetric bias toward |0>. This picture is useful for intuition, though security proofs require full statistical and information-theoretic analysis.

Security warning: The Bloch sphere is a visualization tool, not a security proof. Device imperfections, side channels, finite-key effects, source flaws, and detector attacks require models beyond a single ideal qubit.

17. What the Bloch sphere cannot represent

17.1 It is complete only for one qubit

A general one-qubit density matrix has three independent real parameters, exactly matching a point in the Bloch ball. A two-qubit density matrix has 15 independent real parameters. Two local Bloch vectors provide only six, leaving nine correlation parameters unrepresented.

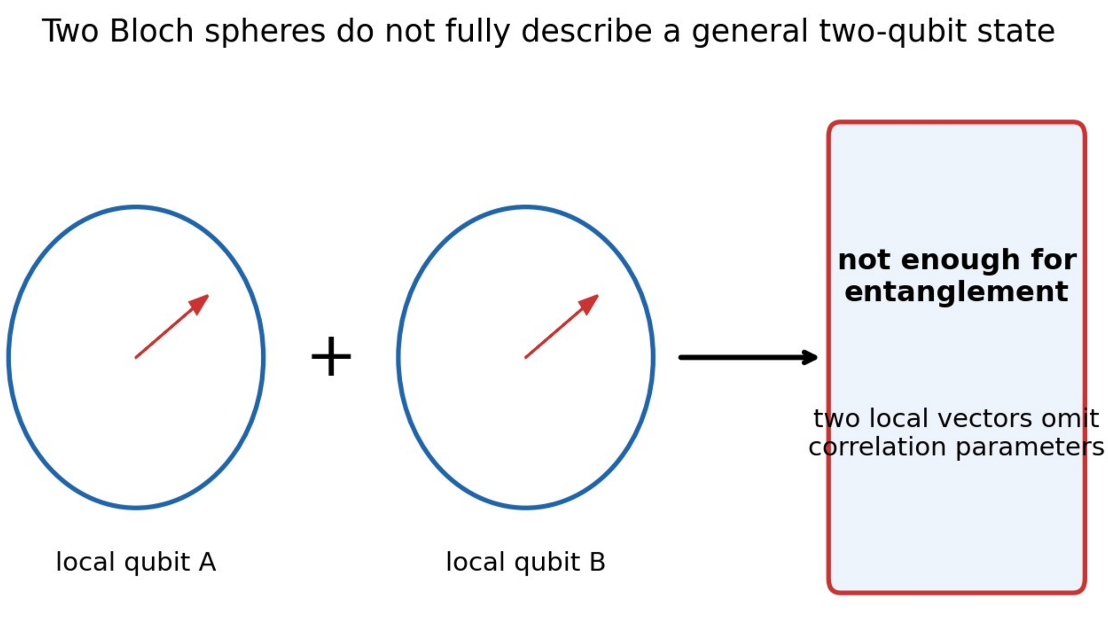

Figure 18. Local Bloch vectors omit two-qubit correlations and entanglement.

17.2 Entanglement can hide behind zero local vectors

For the Bell state (|00>+|11>)/sqrt(2), each individual qubit has reduced state I/2 and therefore local Bloch vector zero. Yet the joint state is pure and maximally entangled. The correlations, not the local vectors, contain the essential information.

$$\rho_{AB} = (1/4)[I\otimes I + X\otimes X - Y\otimes Y + Z\otimes Z]$$

17.3 Generalized Bloch vectors

Higher-dimensional systems can be expanded in a basis of traceless Hermitian generators, producing generalized Bloch vectors. Unlike the qubit case, not every vector inside the naive unit ball corresponds to a valid density matrix, so the geometry becomes substantially more complicated.

17.4 Leakage

A physical qubit is often embedded in a multilevel system. Population leaving the chosen two-level computational subspace cannot be represented by an ordinary Bloch vector. Leakage requires a larger Hilbert-space description or an augmented error model.

Use the sphere responsibly: It is exact for an ideal single qubit, exact for a one-qubit reduced density matrix, and only partial for multi-qubit or multilevel physics.

18. Worked examples

Example 1: Ket to Bloch vector

Given |psi>=(1/2)|0>+(sqrt(3)/2)e^(i pi/3)|1>, identify theta, phi, and r.

Since cos(theta/2)=1/2, theta=2pi/3. The relative phase is phi=pi/3. Therefore

$$r=(\sin(2\pi/3)\cos(\pi/3),\ \sin(2\pi/3)\sin(\pi/3),\ \cos(2\pi/3))$$

$$r=(\sqrt{3}/4,\ 3/4,\ -1/2)$$

Example 2: Bloch vector to density matrix

For r=(0.6,0,0.8),

$$\rho=1/2[[1.8,0.6],[0.6,0.2]]=[[0.9,0.3],[0.3,0.1]]$$

Because |r|=1, the state is pure. One compatible ket is sqrt(0.9)|0>+sqrt(0.1)|1>.

Example 3: Arbitrary-axis measurement

Let r=(0,1,0) and measure along n=(1,1,0)/sqrt(2). Then n dot r=1/sqrt(2).

$$p(+)= (1+1/\sqrt{2})/2, \quad p(-)= (1-1/\sqrt{2})/2$$

Example 4: Gate sequence

Start in |0>, whose vector is +z. Apply H: +z->+x. Apply S: +x->+y. Apply H: +y->-y. The final state is |-i> up to global phase.

Example 5: Dephasing of |+>

The initial vector is (1,0,0). A dephasing channel with eta=0.7 gives r=(0.7,0,0).

$$\rho = [[1/2,0.35],[0.35,1/2]], \quad \text{purity}=(1+0.7^2)/2=0.745$$

Z-basis probabilities remain 1/2, but X-basis visibility falls from 1 to 0.7.

Example 6: Tomography from counts

Suppose 1000 shots per basis produce X counts 700/300, Y counts 450/550, and Z counts 800/200 for +/- outcomes. Then

$$x=0.4, \quad y=-0.1, \quad z=0.6$$

The reconstructed length is sqrt(0.16+0.01+0.36)=sqrt(0.53)≈0.728, so the state is mixed. The density matrix is

$$rho=[[0.8,0.2+0.05i],[0.2-0.05i,0.2]]$$

Example 7: Fidelity of two pure states

Let r=+z and s=+x. Their dot product is zero, so F=(1+0)/2=1/2. This agrees with |<0|+>|^2=1/2.

Example 8: Why local spheres miss entanglement

For |Phi+>=(|00>+|11>)/sqrt(2), tracing out either qubit gives I/2. Both local vectors are zero, identical to two independent maximally mixed qubits. Joint correlations such as <X⊗X>=1 distinguish the entangled state.

Problem-solving habit: Translate the same state among four equivalent descriptions: amplitudes, angles, Bloch vector, and density matrix. Choose whichever makes the requested calculation shortest.

19. Common mistakes and diagnostic checks

Mistake	Why it fails	Correction
Using theta instead of theta/2 in amplitudes	poles and probabilities become wrong	Use cos(theta/2), sin(theta/2)
Treating global phase as a new point	global phase is unobservable for an isolated state	factor out a common phase
Assuming every interior point is pure	mixed states have |r|<1	check purity or vector length
Calling H a 90-degree rotation	its mapping is not a simple quarter-turn	H is pi about (x+z)/sqrt(2)
Using two Bloch spheres for an entangled pair	local vectors omit correlations	use the full density matrix or correlation tensor
Inferring phase from Z-basis counts	Z measurement sees only z	measure X and Y components
Forgetting shot noise	tomography estimates fluctuate	attach uncertainties and enforce physicality

19.1 Fast consistency checks

- For a pure-state Bloch vector, verify x^2+y^2+z^2=1.
- For any density matrix, verify Hermiticity, trace one, and nonnegative eigenvalues.
- Probabilities from any measurement axis must lie in [0,1].
- Unitary gates must preserve |r|.
- Unital noise channels keep r=0 fixed; amplitude damping does not.
- Orthogonal pure states must be antipodal.

19.2 Conversion reference

$$|psi>=cos(theta/2)|0>+e^(i phi)sin(theta/2)|1>$$

r=(sin theta cos phi, sin theta sin phi, cos theta)

rho=(I+r dot sigma)/2

p(+ along n)=(1+n dot r)/2

20. Exercises

A. Core conversions

20. 1. Find the Bloch coordinates of (|0>+i|1>)/sqrt(2).
21. 2. Find a canonical ket for r=(0,-1,0).
22. 3. Convert r=(3/5,4/5,0) to theta and phi.
23. 4. Convert rho=[[3/4,1/4],[1/4,1/4]] to a Bloch vector and determine whether it is pure.
24. 5. Show directly that multiplying a ket by e^(i gamma) does not change x, y, or z.

B. Gates and rotations

25. 6. Starting from r=(1,0,0), find the vector after Z, then H.
26. 7. Apply Ry(pi/2) to |0>. Identify the resulting ket and Bloch point.
27. 8. Show that Rx(theta) leaves the x-coordinate invariant.
28. 9. Find the coordinate action of S^dagger.
29. 10. Determine the Bloch rotation represented by U=exp[-i(pi/3)(X+Z)/(2sqrt(2))].

C. Measurement and tomography

30. 11. A state has r=(0.6,0.8,0). Find probabilities for Z, X, and Y measurements.
31. 12. For r=(0,0,-0.4), measure along n=(1,0,1)/sqrt(2). Find p(+).
32. 13. Tomography gives <X>=0.9, <Y>=0.5, <Z>=0.2. Is the raw estimate physical? Explain.
33. 14. What post-measurement vectors result from an ideal X measurement?
34. 15. How many independent expectation values are required to determine a one-qubit density matrix? Why?

D. Mixed states and noise

35. 16. Compute the purity of r=(0.3,0.4,0).
36. 17. A dephasing channel with eta=0.2 acts on |+i>. Find the output vector and density matrix.
37. 18. A depolarizing channel maps r->0.75r. What happens to purity of an initially pure state?
38. 19. Apply amplitude damping with gamma=1/2 to r=(1,0,0).
39. 20. Explain geometrically why amplitude damping is non-unital.

E. QKD and conceptual reasoning

40. 21. Why does an X-basis measurement of |0> give random outcomes?
41. 22. Place the six-state protocol signals on the sphere.
42. 23. Describe the effect of strong Z dephasing on BB84 X-basis signals.
43. 24. Give two physically different two-qubit states whose local Bloch vectors are both zero.
44. 25. Explain why the Bloch sphere cannot display entanglement completely.

F. Challenge problems

45. 26. Prove |<psi|phi>|^2=(1+r dot s)/2 for pure qubits.
46. 27. Derive the eigenvalues (1+/-|r|)/2 of rho=(I+r dot sigma)/2.
47. 28. Derive the arbitrary-axis measurement formula from the trace rule.
48. 29. Show that conjugation by H maps X to Z, Z to X, and Y to -Y.
49. 30. Starting from the Kraus operators of amplitude damping, derive its affine Bloch transformation.

21. Complete solutions

Solutions 1-5

1. alpha=1/sqrt(2), beta=i/sqrt(2). Thus x=0, y=1, z=0: the +y state |+i>.

2. The -y direction is |-i>=(|0>-i|1>)/sqrt(2).

3. z=0 gives theta=pi/2. atan2(4/5,3/5)=arctan(4/3), so phi≈0.9273 rad.

4. x=2Re(rho01)=1/2, y=-2Im(rho01)=0, z=rho00-rho11=1/2. |r|=1/sqrt(2)<1, so the state is mixed.

5. x and y depend on alpha*beta. Under alpha,beta -> e^(i gamma)alpha,e^(i gamma)beta, the phase cancels: alpha*beta is unchanged. Magnitudes are also unchanged, so z is unchanged.

Solutions 6-10

6. Z maps (1,0,0) to (-1,0,0). H maps (x,y,z)->(z,-y,x), giving (0,0,-1).

7. Ry(pi/2) rotates +z to +x, yielding |+>=(|0>+|1>)/sqrt(2).

8. The three-dimensional x-axis rotation matrix has first row (1,0,0), so x'=x.

9. S^dagger is a -pi/2 z rotation: (x,y,z)->(y,-x,z).

10. The axis is n=(1,0,1)/sqrt(2), and the Bloch rotation angle is pi/3.

Solutions 11-15

11. Z: p(+)=1/2. X: p(+)=(1+0.6)/2=0.8. Y: p(+)=(1+0.8)/2=0.9.

12. n dot r=-0.4/sqrt(2), so p(+)=(1-0.4/sqrt(2))/2≈0.3586.

13. The length is sqrt(0.81+0.25+0.04)=sqrt(1.10)>1, so it is unphysical. Finite-shot error or calibration error produced a vector outside the Bloch ball.

14. The output vectors are +x for outcome + and -x for outcome -.

15. Three real values. A 2x2 Hermitian matrix has four real parameters, and trace one removes one.

Solutions 16-20

16. |r|^2=0.09+0.16=0.25, so purity=(1+0.25)/2=0.625.

17. |+i> has r=(0,1,0). Dephasing gives (0,0.2,0), so rho=[[1/2,-0.1i],[0.1i,1/2]].

18. The output length is 0.75, so purity=(1+0.75^2)/2=0.78125.

19. x -> sqrt(1/2), y=0, z -> 1/2. The output is (1/sqrt(2),0,1/2).

20. A unital channel maps the center r=0 to itself. Amplitude damping maps it to (0,0,gamma), translating the center toward |0>.

Solutions 21-25

21. The vectors +z and +/-x are perpendicular, so their dot product is zero and both X outcomes have probability 1/2.

22. Use +/-x, +/-y, and +/-z: the six Pauli eigenstates.

23. Strong Z dephasing shrinks x and y toward zero. The X-basis signals become difficult to distinguish and approach I/2.

24. The Bell state |Phi+> and the product mixed state I/2⊗I/2 both have zero local Bloch vectors, but their joint correlations are completely different.

25. A general two-qubit state needs 15 real parameters. Two local vectors supply only six and omit nine correlation components, including entanglement information.

Solution 26

Write the pure-state density matrices as rho=(I+r dot sigma)/2 and sigma=(I+s dot sigma)/2. Since Tr(rho sigma)=|<psi|phi>|^2, expand the product. Use Tr(I)=2, Tr(sigma_i)=0, and Tr(sigma_i sigma_j)=2 delta_ij. The result is (1+r dot s)/2.

Solution 27

Because (r dot sigma)^2=|r|^2 I, the operator r dot sigma has eigenvalues +/-|r|. Adding I and dividing by two gives eigenvalues (1+/-|r|)/2.

Solution 28

Use P_+=(I+n dot sigma)/2 and rho=(I+r dot sigma)/2. Then p(+)=Tr(P_+rho). Expand and use the Pauli trace identities. Only Tr(I)=2 and Tr[(n dot sigma)(r dot sigma)]=2 n dot r survive, yielding (1+n dot r)/2.

Solution 29

Direct matrix multiplication gives HXH=Z and HZH=X. Since Y=iXZ, conjugation gives HYH=i(HXH)(HZH)=iZX=-Y.

Solution 30

For amplitude damping, E0=[[1,0],[0,sqrt(1-gamma)]] and E1=[[0,sqrt(gamma)],[0,0]]. Evaluate rho'=E0 rho E0^dagger+E1 rho E1^dagger for rho=[[(1+z)/2,(x-iy)/2],[(x+iy)/2,(1-z)/2]]. Reading the output entries gives x'=sqrt(1-gamma)x, y'=sqrt(1-gamma)y, and z'=(1-gamma)z+gamma.

22. Compact reference sheet

Concept	Formula
Pure state	|psi>=cos(theta/2)|0>+e^(i phi)sin(theta/2)|1>
Bloch coordinates	(sin theta cos phi, sin theta sin phi, cos theta)
From amplitudes	x=2Re(alpha*beta), y=2Im(alpha*beta), z=|alpha|^2-|beta|^2
Density matrix	rho=(I+r dot sigma)/2
Purity	Tr(rho^2)=(1+|r|^2)/2
Measurement	p(+ along n)=(1+n dot r)/2
Rotation gate	U_n(theta)=exp[-i theta(n dot sigma)/2]
Pure-state fidelity	F=(1+r dot s)/2
Trace distance	D=|r-s|/2
General channel	r->Ar+c

23. Final perspective

The Bloch sphere unifies much of single-qubit quantum mechanics. A ket becomes a direction. Relative phase becomes azimuth. Computational-basis bias becomes height. Pauli expectations become Cartesian coordinates. Unitary gates become rotations. Pure-state overlap becomes angular separation. Mixedness becomes radial contraction. Measurement becomes projection onto an axis, and common noise processes become geometric deformations of the ball.

Its greatest strength is also its limitation: the picture is complete because one-qubit density matrices have exactly three independent real parameters. As soon as several qubits are combined, correlations and entanglement require a larger mathematical description. The sphere should therefore be used as a precise one-qubit model and as a local diagnostic inside multi-qubit systems, not as a replacement for full Hilbert-space reasoning.

Mastery checkpoint: You should now be able to move fluently among ket, angle, Bloch-vector, density-matrix, and measurement descriptions; predict single-qubit gate motion; reconstruct states from Pauli data; and diagnose basic noise geometrically.

Recommended next step

Topic 14 develops practical Qiskit skills: constructing circuits, simulating statevectors and measurement counts, visualizing states, transpiling for hardware, and connecting the mathematical objects in these volumes to executable quantum programs.

Selected references for deeper study

- M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information.
- J. Preskill, Lecture Notes on Quantum Computation.
- J. J. Sakurai and J. Napolitano, Modern Quantum Mechanics (spin-1/2 geometry).
- IBM Quantum Learning materials on single-qubit states, operators, and measurement.